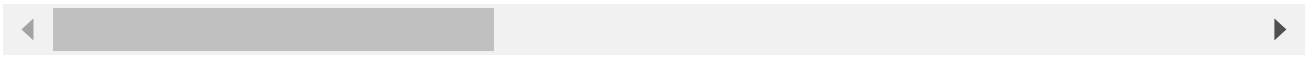
Inport dataset

```
1  import pandas as pd
2  Df = pd.read_csv("/content/housesalesprediction.zip")
3  Df
```

	id	date	price	bedrooms	bathrooms	sqft_living	sqft
0	7129300520	20141013T000000	221900.0	3	1.00	1180	
1	6414100192	20141209T000000	538000.0	3	2.25	2570	
2	5631500400	20150225T000000	180000.0	2	1.00	770	1
3	2487200875	20141209T000000	604000.0	4	3.00	1960	
4	1954400510	20150218T000000	510000.0	3	2.00	1680	
...	...	...	...	...	...	...	
21608	263000018	20140521T000000	360000.0	3	2.50	1530	
21609	6600060120	20150223T000000	400000.0	4	2.50	2310	
21610	1523300141	20140623T000000	402101.0	2	0.75	1020	
21611	291310100	20150116T000000	400000.0	3	2.50	1600	
21612	1523300157	20141015T000000	325000.0	2	0.75	1020	

21613 rows × 21 columns





```
1 Df.groupby(["floors","grade"]).first()
```

		id	date	price	bedrooms	bathrooms	sqft_liv
floors	**grade**						
1.0	**1**	3980300371	20140926T000000	142000.0	0	0.00	
	3	3421079032	20150217T000000	75000.0	1	0.00	
	4	7325600160	20140604T000000	299000.0	1	0.75	
	5	9435300030	20140528T000000	550000.0	4	1.00	1
	6	5631500400	20150225T000000	180000.0	2	1.00	
	7	7129300520	20141013T000000	221900.0	3	1.00	1
	8	1954400510	20150218T000000	510000.0	3	2.00	1
	9	2524049179	20140826T000000	2000000.0	3	2.75	3
	10	825059270	20141121T000000	1095000.0	5	3.00	4
	11	7237550310	20140512T000000	1225000.0	4	4.50	5
	12	6838800140	20150224T000000	1100000.0	4	3.50	4
	13	853200010	20140701T000000	3800000.0	5	5.50	7
1.5	**4**	3089000005	20140724T000000	150000.0	2	1.00	
	5	9349900105	20150407T000000	795000.0	2	1.00	1
	6	1202000200	20141103T000000	233000.0	3	2.00	1
	7	114101516	20140528T000000	310000.0	3	1.00	1
	8	3303700376	20141201T000000	667000.0	3	1.00	1
	9	1952200240	20140611T000000	850830.0	3	2.50	2
	10	1954700695	20140612T000000	2250000.0	5	4.25	4
	11	7960900060	20150504T000000	2900000.0	4	3.25	5
2.0	**5**	423059184	20141201T000000	180000.0	3	1.00	1
	6	2172000075	20140623T000000	290900.0	2	2.00	1
	7	6414100192	20141209T000000	538000.0	3	2.25	2
	8	7137970340	20140703T000000	285000.0	5	2.50	2
	9	9297300055	20150124T000000	650000.0	4	3.00	2
	10	3394100030	20140909T000000	975000.0	4	2.50	2
	11	4054500390	20141007T000000	1365000.0	4	4.75	5
	12	7855801670	20150401T000000	2250000.0	4	3.25	5
	13	2470100110	20140804T000000	5570000.0	5	5.75	9
2.5	**6**	4013800131	20140807T000000	267500.0	2	1.00	1
	7	7234600796	20141126T000000	495000.0	2	1.75	1
	8	770000045	20141024T000000	405600.0	5	1.50	2

	8	7700000045	20141024T000000	405000.0	5	1.50	2
	9	2625069070	20150410T000000	1385000.0	4	3.25	4
	10	9103000393	20140910T000000	1225000.0	5	4.50	3
	11	7524900003	20141210T000000	3278000.0	2	1.75	6
	12	5249800010	20141203T000000	2725000.0	4	4.25	6
	13	9831200500	20150304T000000	2479000.0	5	3.75	6
3.0	7	2426039314	20141201T000000	280000.0	2	1.50	1

```
vector_product = pd.DataFrame({'grade' : Df['grade'] ,
                               'price' : Df['price'] ,
                               'floors': Df['floors']})
vector_product
```

	grade	price	floors
0	7	221900.0	1.0
1	7	538000.0	2.0
2	6	180000.0	1.0
3	7	604000.0	1.0
4	8	510000.0	1.0
...	...	...	...
21608	8	360000.0	3.0
21609	8	400000.0	2.0
21610	7	402101.0	2.0
21611	8	400000.0	2.0
21612	7	325000.0	2.0

21613 rows × 3 columns

```
vector_sum = pd.DataFrame({'price' : Df['price'] ,
                           'grade' : Df['grade'] ,
                           'floors': Df['floors']})
vector_sum
```

	price	grade	floors
0	221900.0	7	1.0
1	538000.0	7	2.0
2	180000.0	6	1.0
3	604000.0	7	1.0
4	510000.0	8	1.0
...	...	...	...
21608	360000.0	8	3.0
21609	400000.0	8	2.0
21610	402101.0	7	2.0
21611	400000.0	8	2.0
21612	325000.0	7	2.0

21613 rows × 3 columns

✓ 0s completed at 8:05 PM